Effat University

CS4082 - Machine Learning

Health Risk Prediction Using Machine Learning **Across Heterogeneous Patient Populations**

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Motivation

- *ICU decisions must be made quickly under pressure*
- *Incorrect predictions can affect patient outcomes*
- *Patient data is complex and varies across cases*

Problem

Problem Context	*Real-Word Issue*
ICU mortality prediction relies on complex clinical data	*ICU data is distributed across multiple tables*
Multiple features must be combined for accurate modeling	*Missing values and variability increase modeling difficulty*

Objective

01 *Predict ICU patient mortality using classification models*

02 *Combine clinical features into a unified input*

03 *Prepare and structure data for modeling*

04 *Address missing values and data variability*

05 *Evaluate models using performance metrics*

Dataset

- *eICU Collaborative Research Database (PhysioNet)*
- *Multi-center ICU dataset with de-identified patient records*
- *Contains vital signs, diagnoses, treatments, and severity scores (e.g., APACHE)*
- *Includes data from thousands of ICU stays across multiple hospitals (only a subset were used)*
- *Final prepared dataset contained 2,520 ICU patient records.*
- *Target was imbalanced: 2,308 survived and 212 died.*
- *This imbalance made mortality prediction harder because the model saw far fewer positive mortality cases.*

Pipeline

Table Selection

- *patient*
- *apachePatientResult*
- *apachePredVar*
- *apacheApsVar*

Variable Extraction

- *demographics*
- *clinical indicators*
- *vital signs*

Table Merging

- *using patientunitstayid*

Dataset Creation

- *aggregate to one row per patient*

Cleaning & Missing Values

- *imputation*
- *data consistency checks*

Feature Selection

- *selected relevant clinical predictors and tested SelectKBest.*

Model Training

- *used Python/scikit-learn pipelines with preprocessing, baseline and tuned models.*

Model Evaluation

- *evaluated validation and test results using accuracy, precision, recall, F1-score, and AUC-ROC.*

Method

Approach

- *Binary classification (ICU mortality prediction)*
- *Input: processed clinical features*
- *Output: patient outcome (alive / deceased)*
- *Data preprocessing and feature engineering applied before modeling*

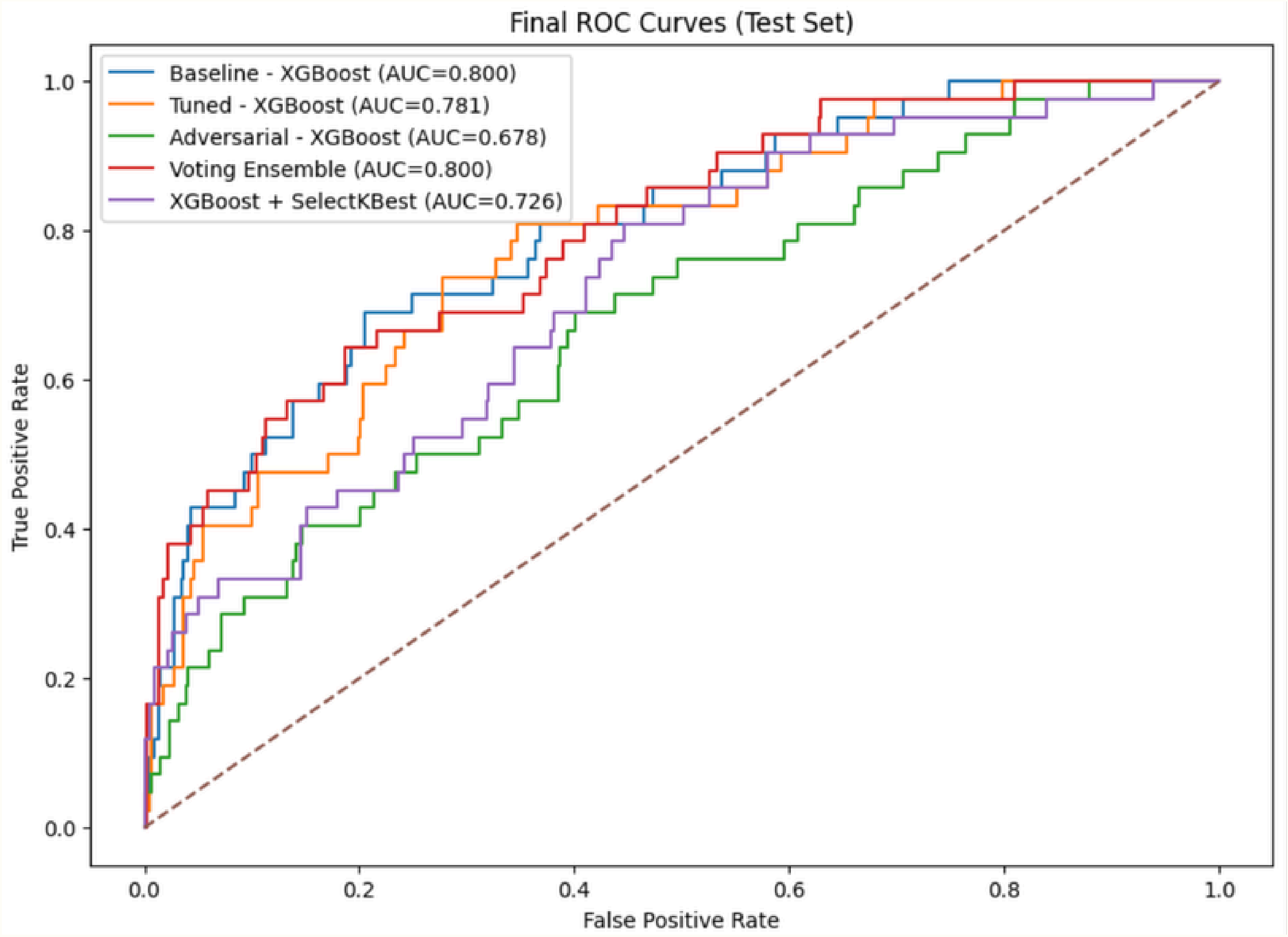
Models Tested

- *Logistic Regression, Random Forest, Decision Tree*
- *KNN, SVM, Naive Bayes*
- *Gradient Boosting, XGBoost*
- *Models implemented using scikit-learn and XGBoost libraries*

Model Selection, Tuning & Ensemble

- *Selected top-performing models: Logistic Regression, XGBoost, Gradient Boosting*
- *Hyperparameter tuning performed using GridSearchCV with 5-fold cross-validation*
- *Key tuned parameters:*
  - *XGBoost: learning rate, max depth, number of estimators*
  - *Gradient Boosting: learning rate, number of estimators, max depth*
  - *Logistic Regression: regularization strength (C)*
- *Final model: Voting Ensemble*
- *Ensemble improved performance, especially for the minority class (higher F1-score)*

Evalutation & Results



	Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
3	Voting Ensemble	0.892857	0.380000	0.452381	0.413043	0.800041
0	Baseline - XGBoost	0.797619	0.227273	0.595238	0.328947	0.799732
1	Tuned - XGBoost	0.861111	0.274194	0.404762	0.326923	0.780561
4	XGBoost + SelectKBest	0.815476	0.200000	0.404762	0.267717	0.725675
2	Adversarial - XGBoost	0.561508	0.125523	0.714286	0.213523	0.678056

- *XGBoost was selected as a strong base model and used in the final ensemble*
- *Improving performance on the minority class was challenging*
- *Initial F1-score was very low (~0.1)*
- *Applied hyperparameter tuning and ensemble learning to improve performance*
- *Tested adversarial training by introducing controlled noise/perturbations to improve model robustness*
- *Performance improved, but class imbalance remains a limitation*
- *High accuracy with lower F1-score indicates class imbalance*

Reflection & Future Work

Reflection

- *Class imbalance significantly impacted model performance*
- *Accuracy was misleading because the dataset was imbalanced; F1-score was more informative because it combines precision and recall for the minority mortality class.*
- *Real-world clinical data is complex and noisy*
- *Model improvements required multiple strategies*

Future Work

- *Apply class balancing techniques (e.g., SMOTE, class weighting)*
- *Improve feature engineering for better signal extraction*
- *Explore more advanced ensemble methods*
- *Focus on improving minority class prediction*

Conclusion

- Developed an ICU mortality prediction pipeline using clinical data
- Compared multiple models and identified top performers
- Voting Ensemble achieved the best overall performance
- Model performance improved through tuning and model selection
- Results highlight challenges of predicting the minority class

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Thank You
For Listening!

Any Questions?

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